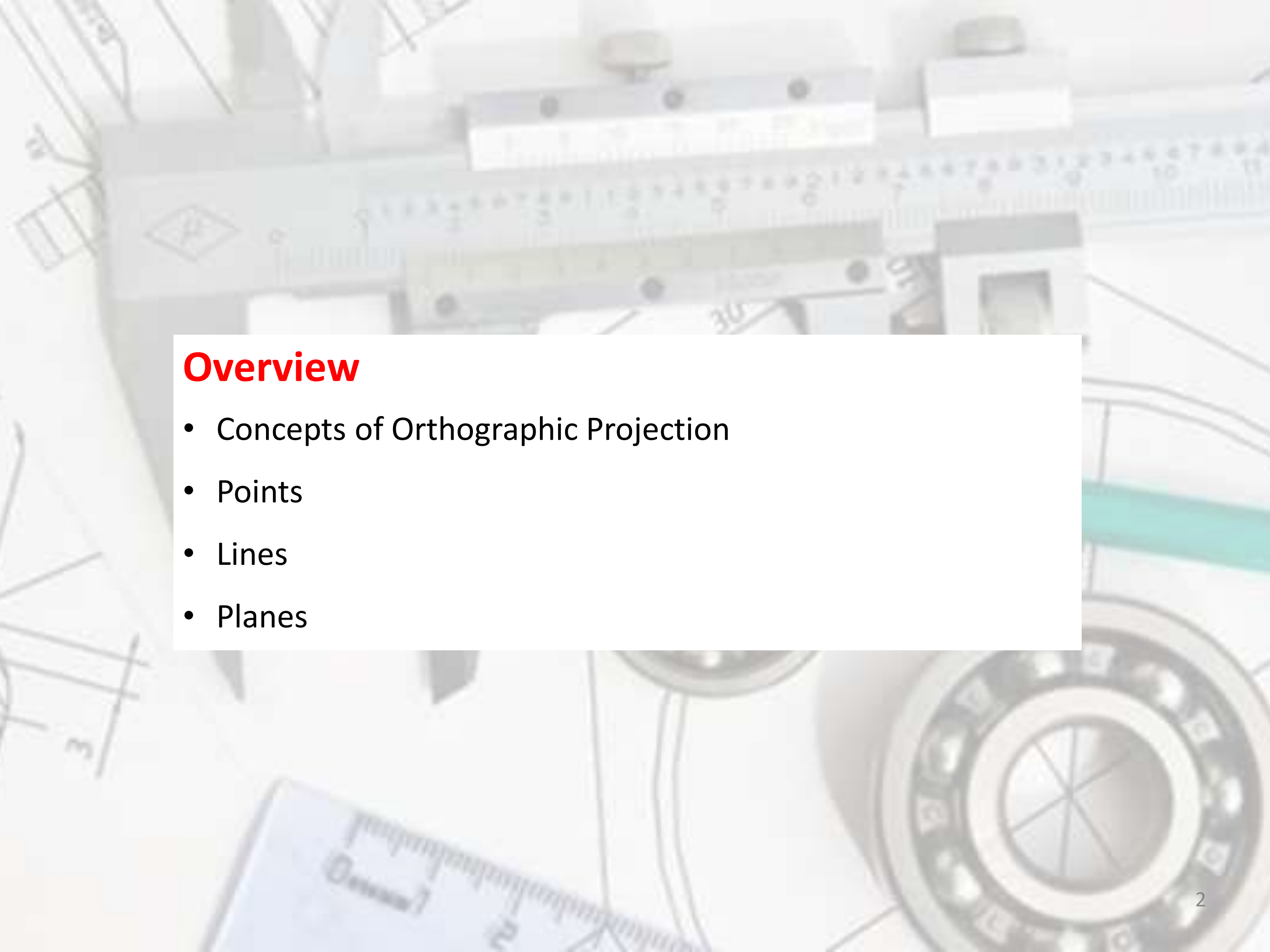


ENGINEERING DRAWING- LECTURE 1

POINTS, LINES, AND PLANES

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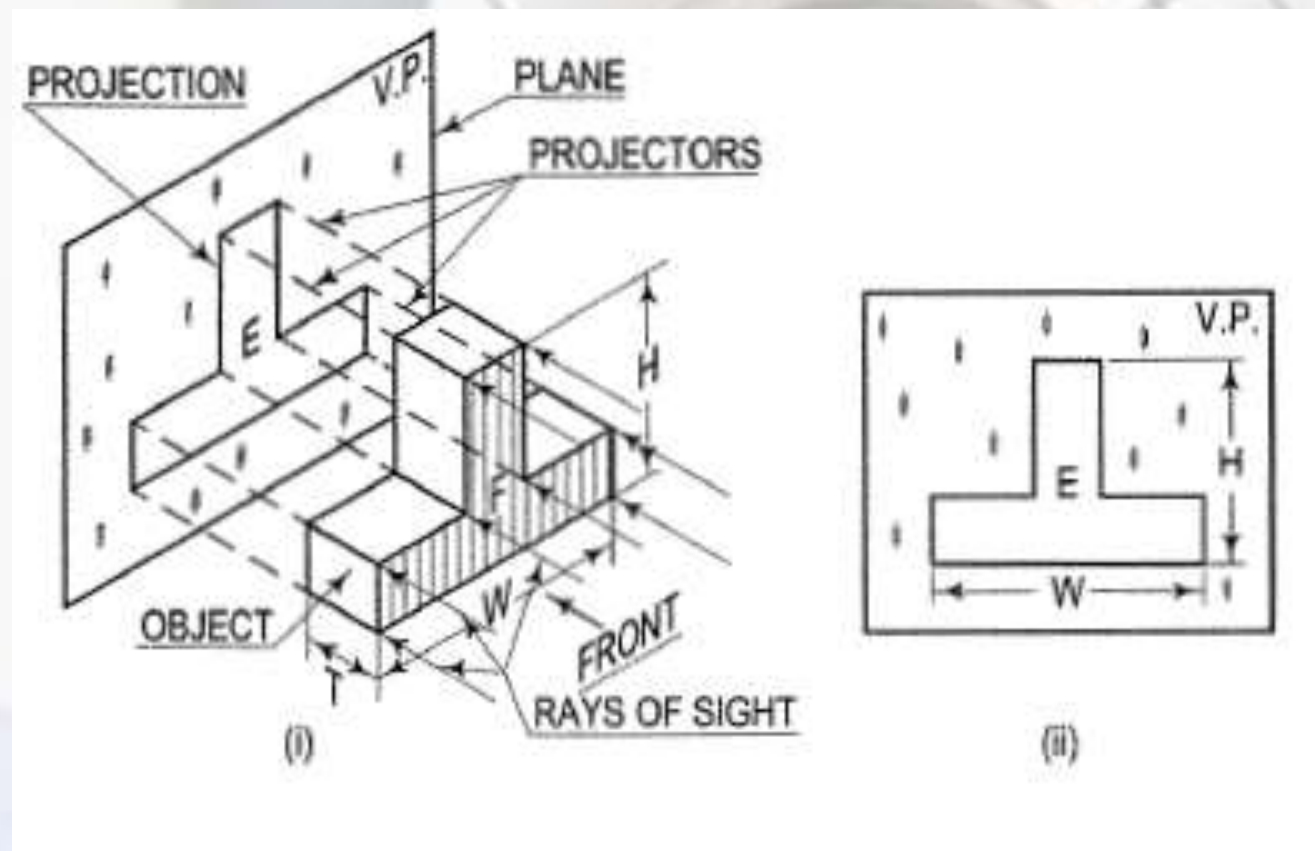


Overview

- Concepts of Orthographic Projection
- Points
- Lines
- Planes

Concepts of Orthographic Projection

- If straight lines are drawn from various points on the contour of an object to meet a plane, the object is said to be **projected** on that plane
- The figure formed by joining in correct sequence the points at which these lines meet the plane is called the **projection** of the object
- The line from the object to the plane is called **projectors**

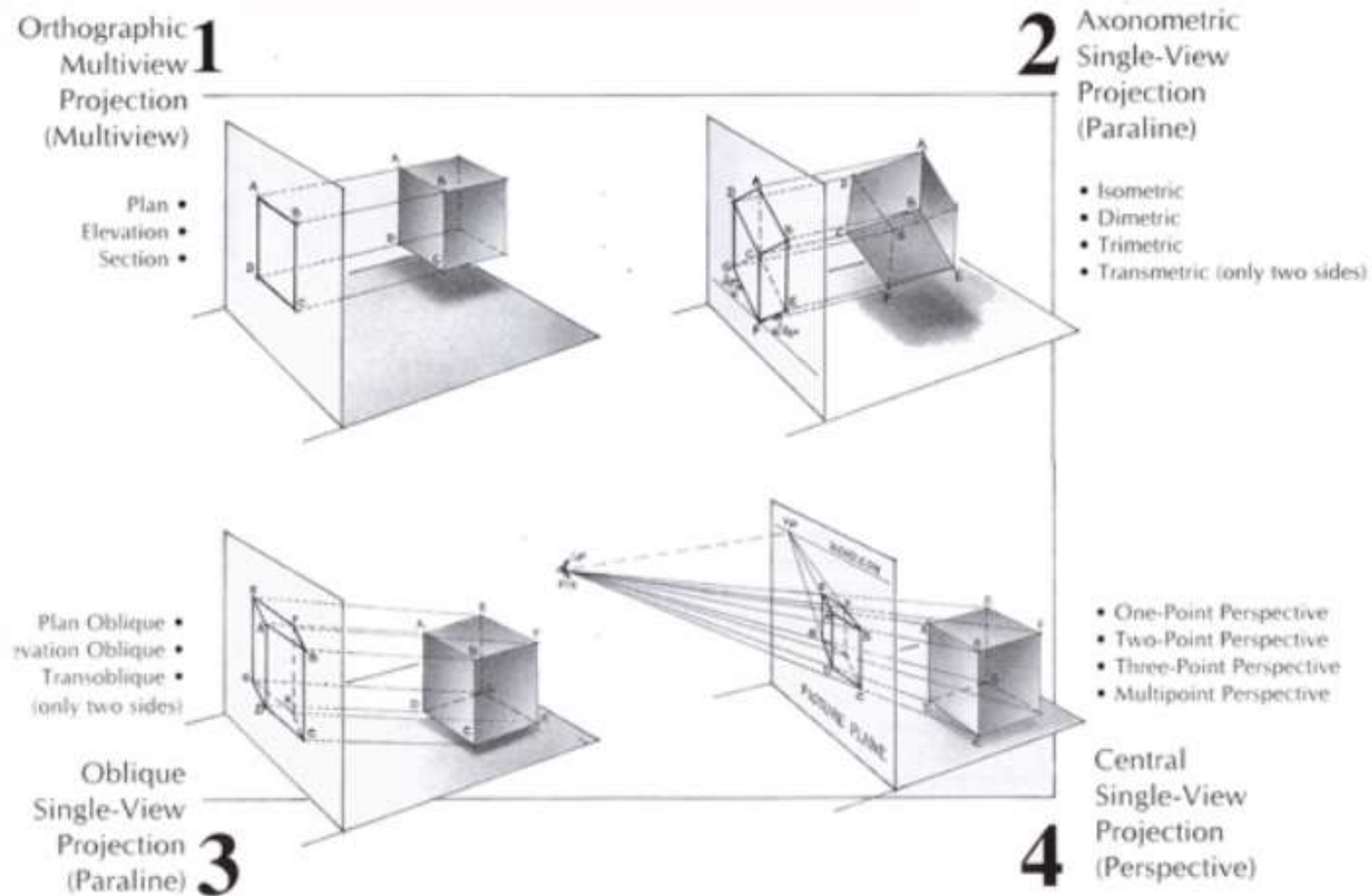


Source: N. D. Bhatt

Concepts of Orthographic Projection

- Methods of Projection

i. Orthographic ii. Isometric iii. Oblique iv. Perspective

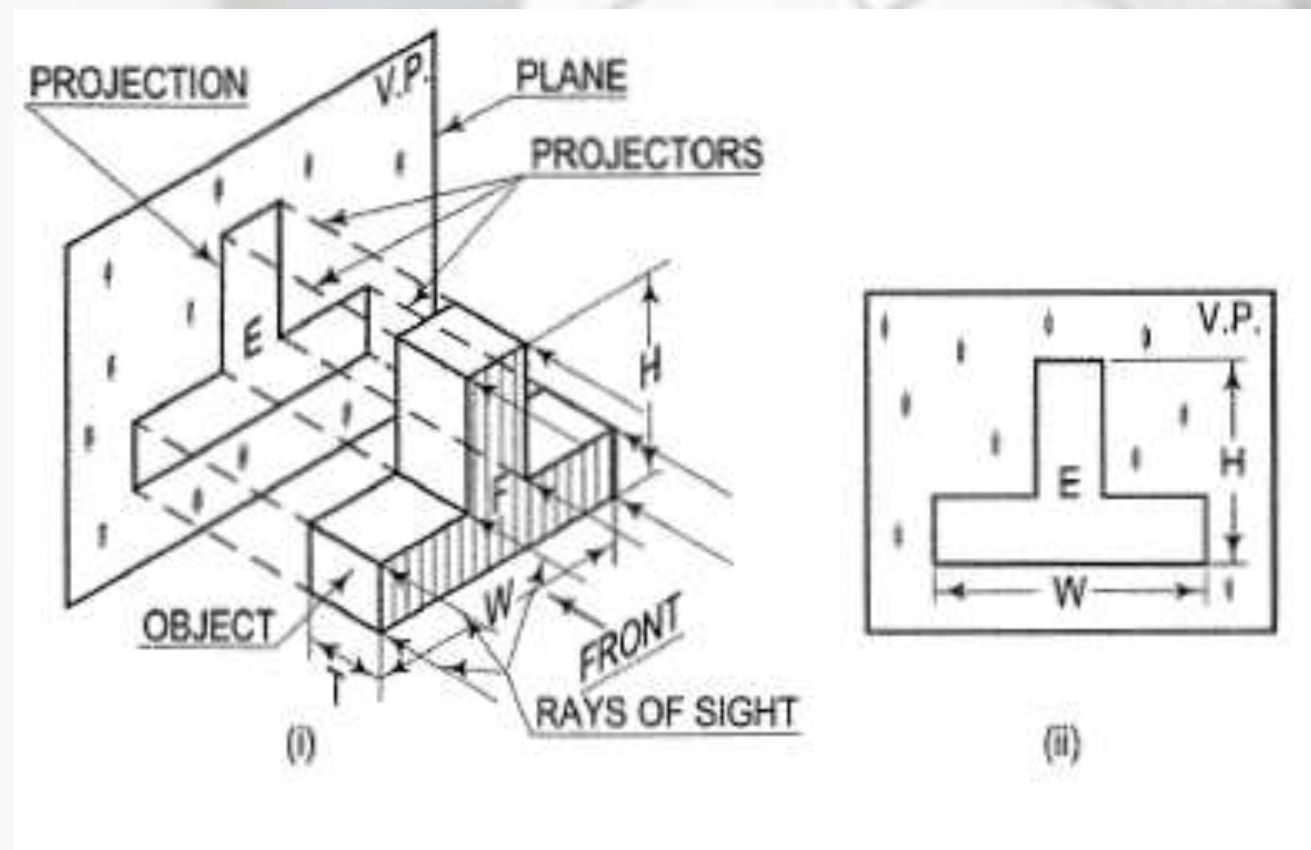


Source: N. D. Bhatt

http://studiomaven.org/Context__Overview_of_Architectural_Drawings.html

Concepts of Orthographic Projection contd.

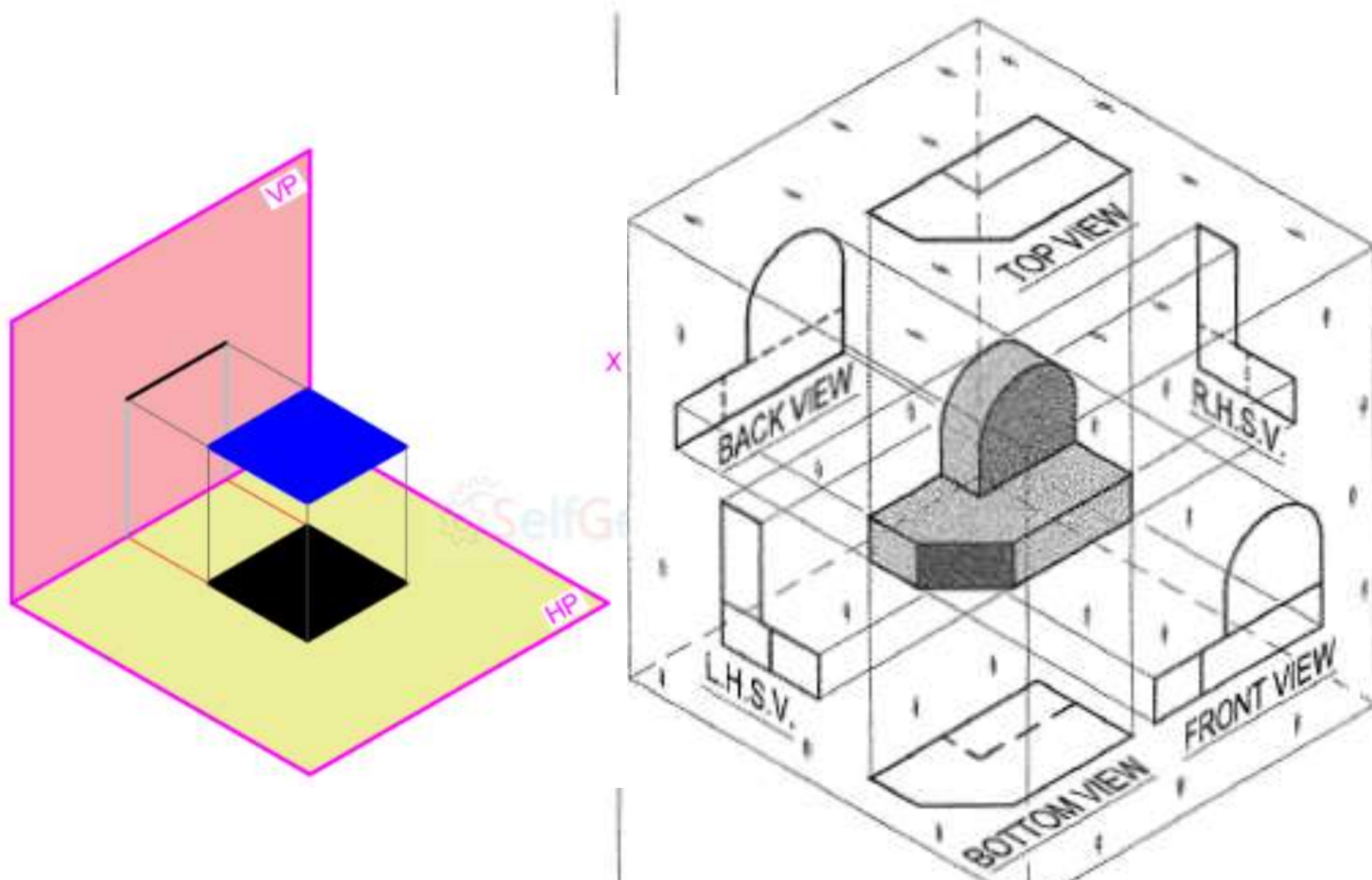
- When the projectors are parallel to each other and also perpendicular to the plane, the projection is called **orthographic projection**



Concepts of Orthographic Projection contd.

- In other words, orthographic projections is a technical drawing in which different views of an object are projected on different reference planes observing perpendicular to respective reference plane
- **Different reference planes are:**
 - *Horizontal Plane (HP)*
 - *Vertical Plane (VP)*
 - *Side or Profile Plane (PP)*
- **Different views are:**
 - *Front View (FV)*
 - *Top View (TV)*
 - *Side View (RHSV, LHSV)*

Concepts of Orthographic Projection contd.



TERMS 'ABOVE' & 'BELOW' WITH RESPECT TO H.P.
AND TERMS 'INFRONT' & 'BEHIND' WITH RESPECT TO V.P.

Sources: N. D. Bhatt

<https://www.selfgeared.com/projection-of-plane-in-engineering-drawing/>

Concepts of Orthographic Projection contd.

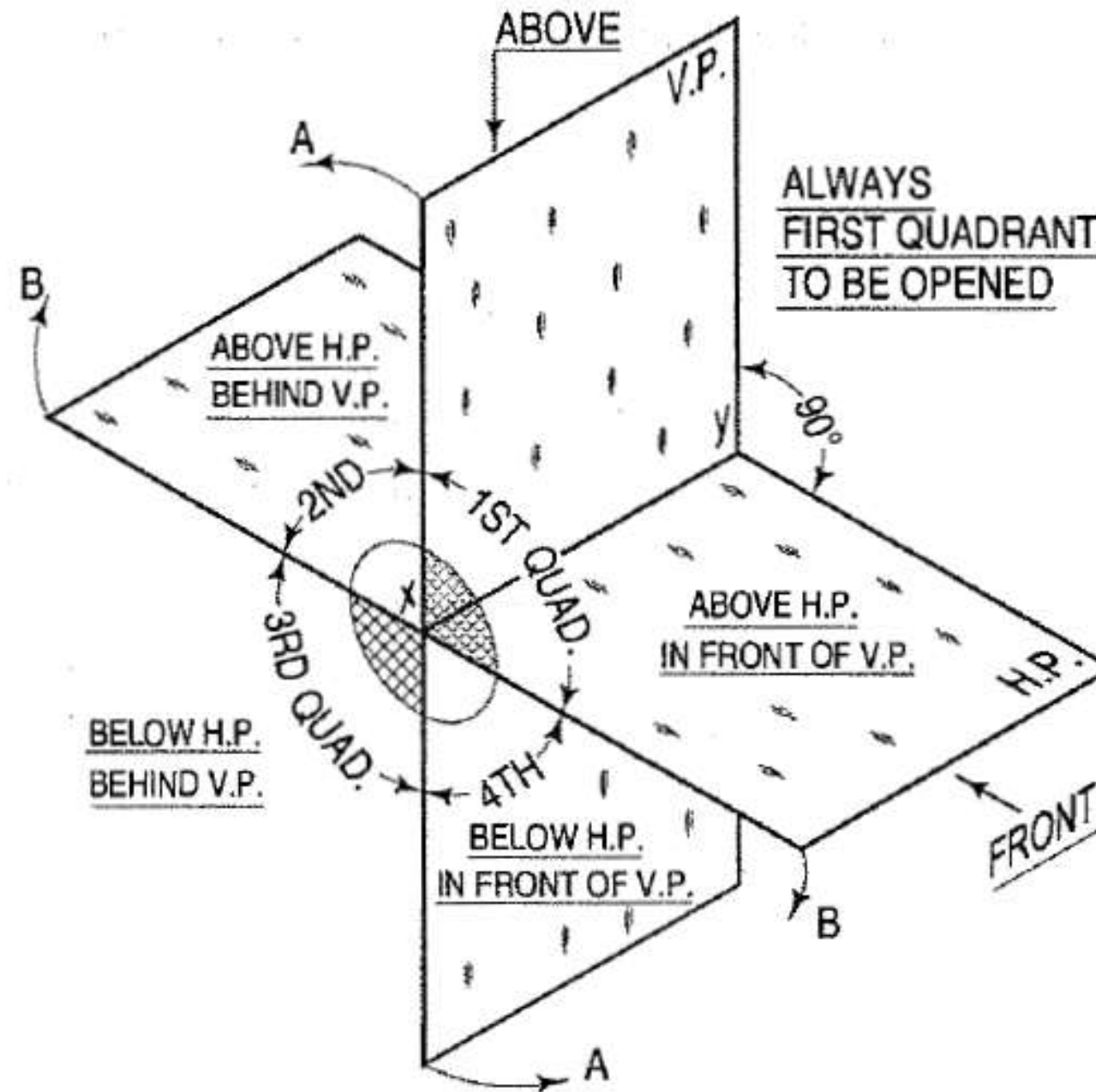
NOTATIONS

Following notations should be followed while naming Different views in orthographic projections.

OBJECT	POINT A	LINE AB
IT'S TOP VIEW	a	a b
IT'S FRONT VIEW	a'	a' b'
IT'S SIDE VIEW	a''	a'' b''

Concepts of Orthographic Projection contd.

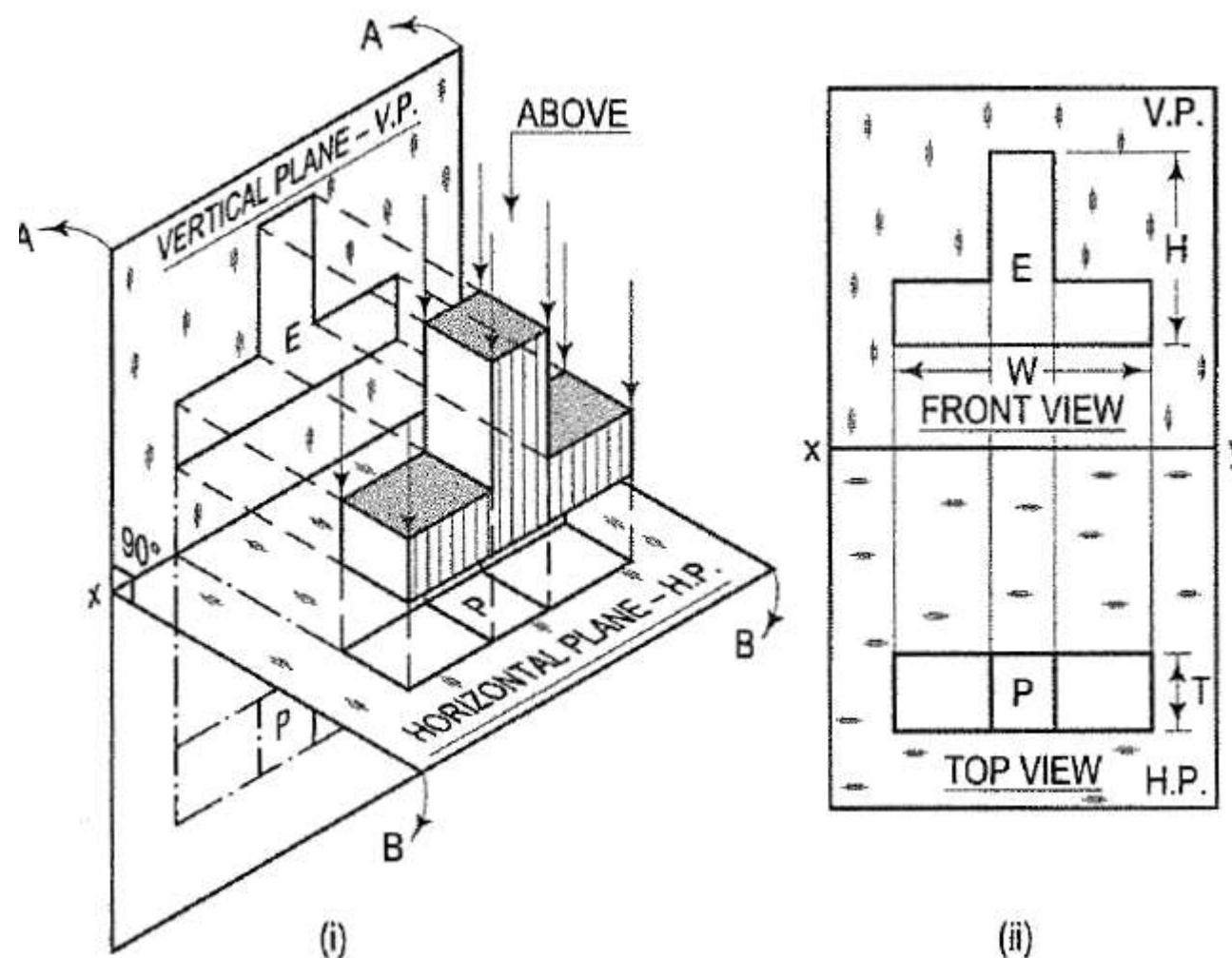
- Four Quadrants



Source: N. D. Bhatt

Concepts of Orthographic Projection contd.

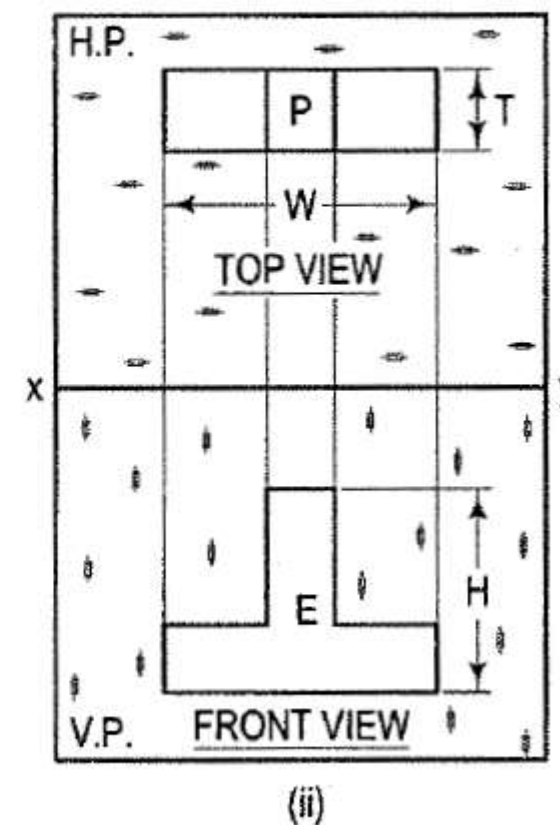
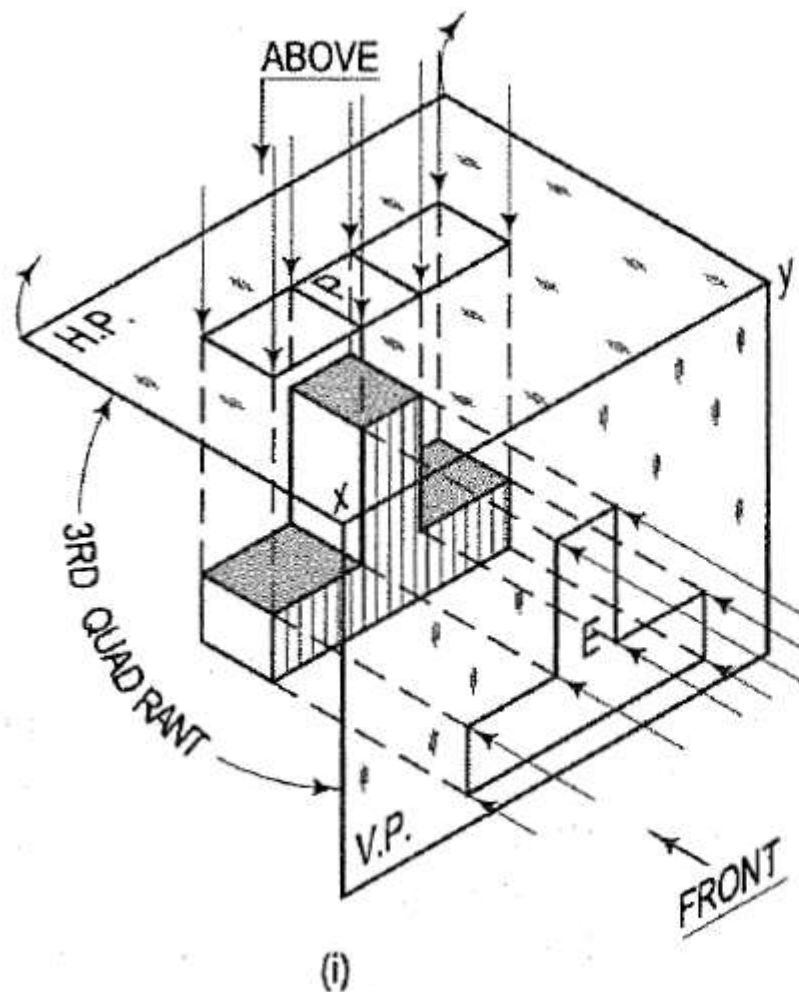
- First Angle Projection
- The object is assumed to be situated in front of the VP and above the HP; i.e., in the first quadrant and then projected it on these planes. This method of projection is known as first-angle projection method



Source: N. D. Bhatt

Concepts of Orthographic Projection contd.

- Third Angle Projection
- The object is assumed to be situated in the third quadrant. The planes of projection are assumed to be transparent

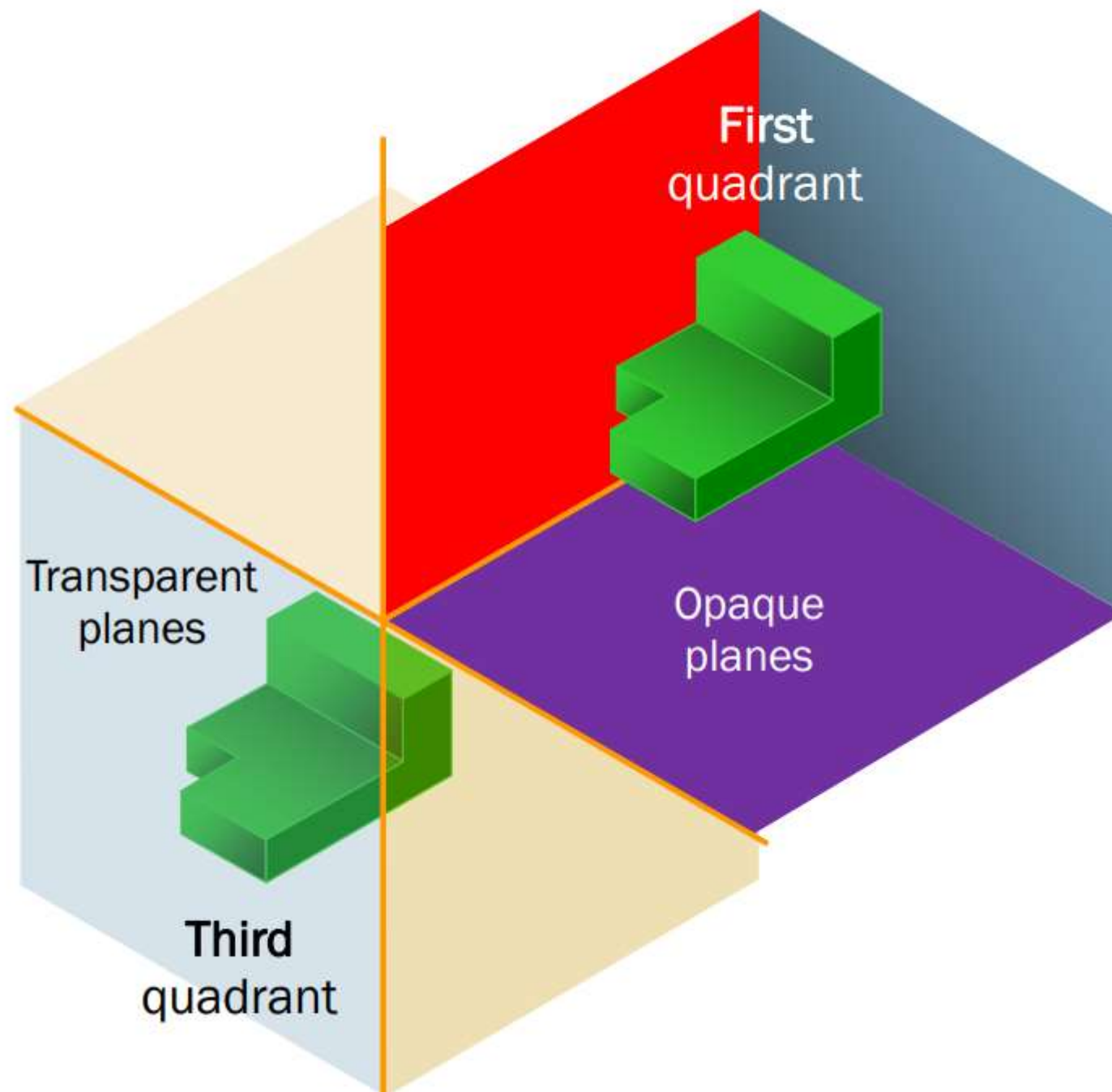


Concepts of Orthographic Projection contd.

DIFFERENCE BETWEEN FIRST-ANGLE PROJECTION METHOD AND THIRD-ANGLE PROJECTION METHOD

No.	First-angle projection method	Third-angle projection method
1.	The object is kept in the <i>first quadrant</i> .	The object is assumed to be kept in the <i>third quadrant</i> .
2.	The object lies between the observer and the plane of projection.	The plane of projection lies between the observer and the object.
3.	The plane of projection is assumed to be non-transparent.	The plane of projection is assumed to be <i>transparent</i> .
4.	In this method, when the views are drawn in their relative positions, the <i>plan</i> comes <i>below</i> the <i>elevation</i> , the view of the object as observed from the <i>left-side</i> is drawn to the <i>right of elevation</i> .	In this method, when the views are drawn in their relative positions, the <i>plan</i> , comes <i>above</i> the <i>elevation</i> , <i>left hand side view</i> is drawn to the <i>left hand side of the elevation</i> .
5.	This method of projection is now recommended by the "Bureau of Indian Standards" from 1991.	This method of projection is used in U.S.A. and also in other countries.

Concepts of Orthographic Projection contd.



Source:

[https://www.iitg.ac.in/kpmech/ME111-2016/ORTHOGRAPHIC%20PROJECTIONS-1%20\(2016\).pdf](https://www.iitg.ac.in/kpmech/ME111-2016/ORTHOGRAPHIC%20PROJECTIONS-1%20(2016).pdf)

Projections of Points

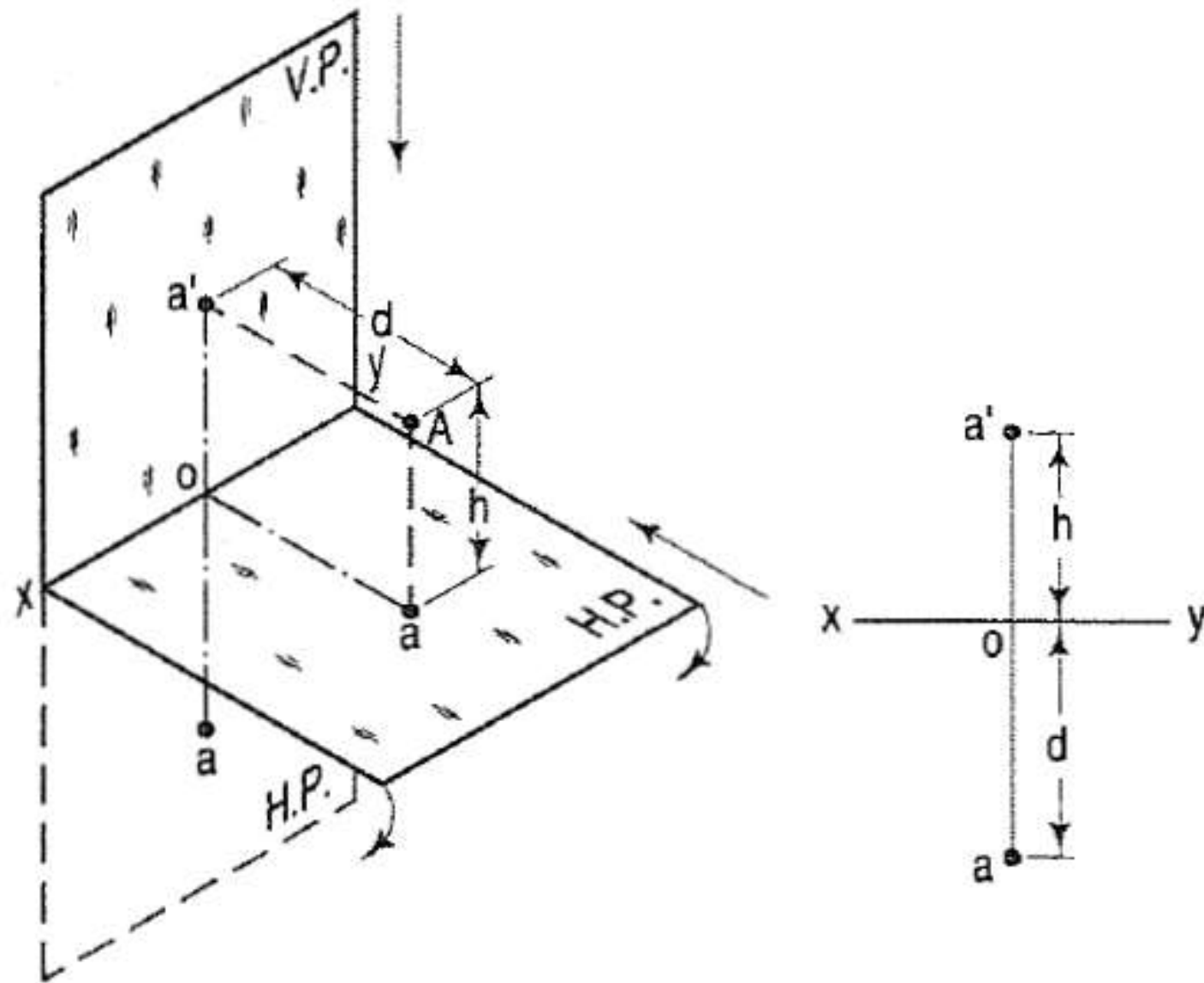
- Introduction
- Point Situated in First Quadrant
- Point Situated in Second Quadrant
- Point Situated in Third Quadrant
- Practice Problem

Projections of Points contd.

- A point may be situated in space, in any one of the four quadrants formed by the two principal planes of projection or may lie in any one or both of them

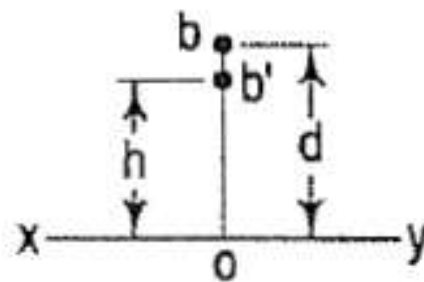
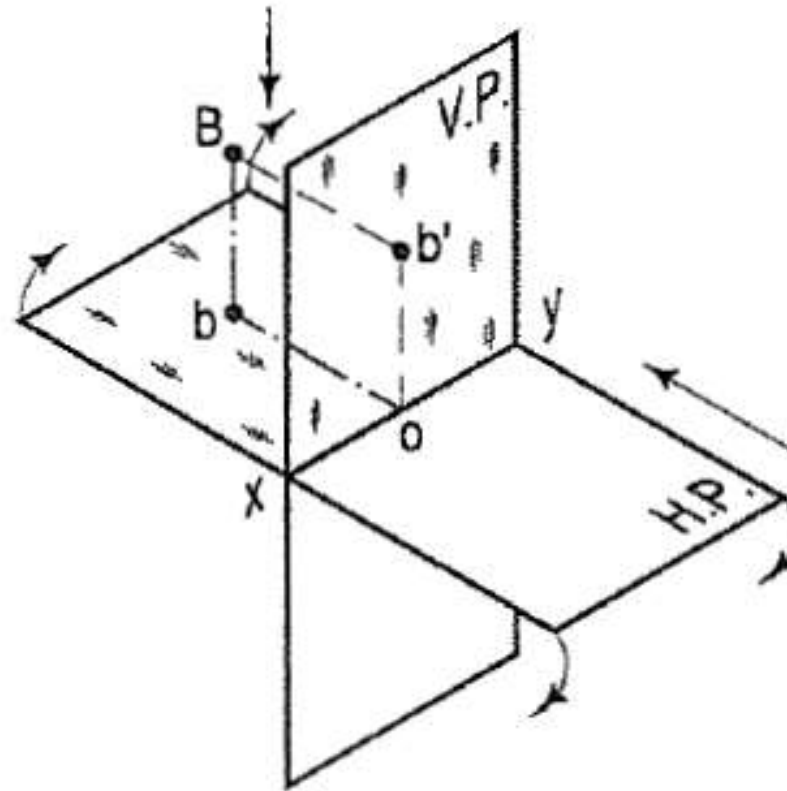
Projections of Points contd.

- 1. Point Situated in the first Quadrant



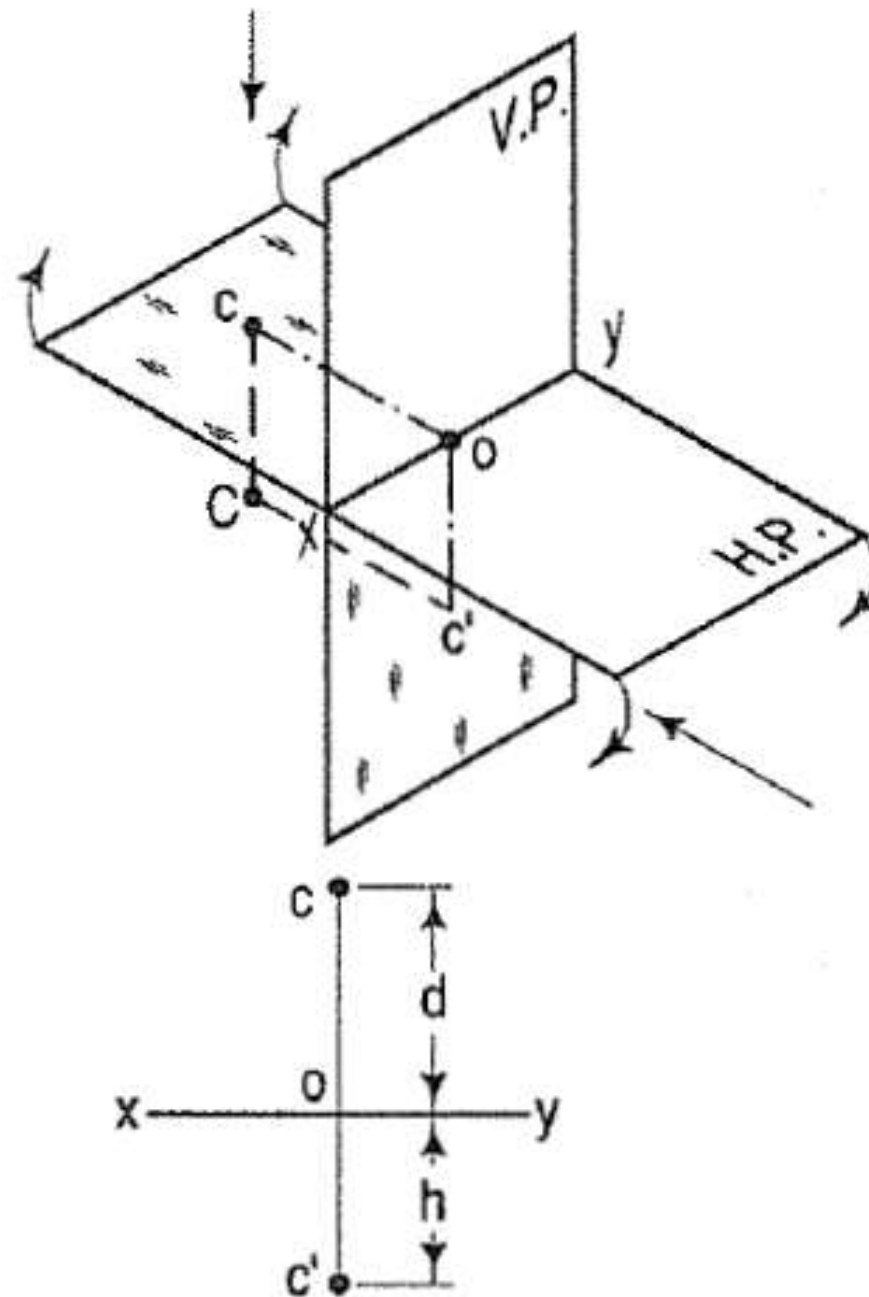
Projections of Points contd.

- 2. Point Situated in the second Quadrant



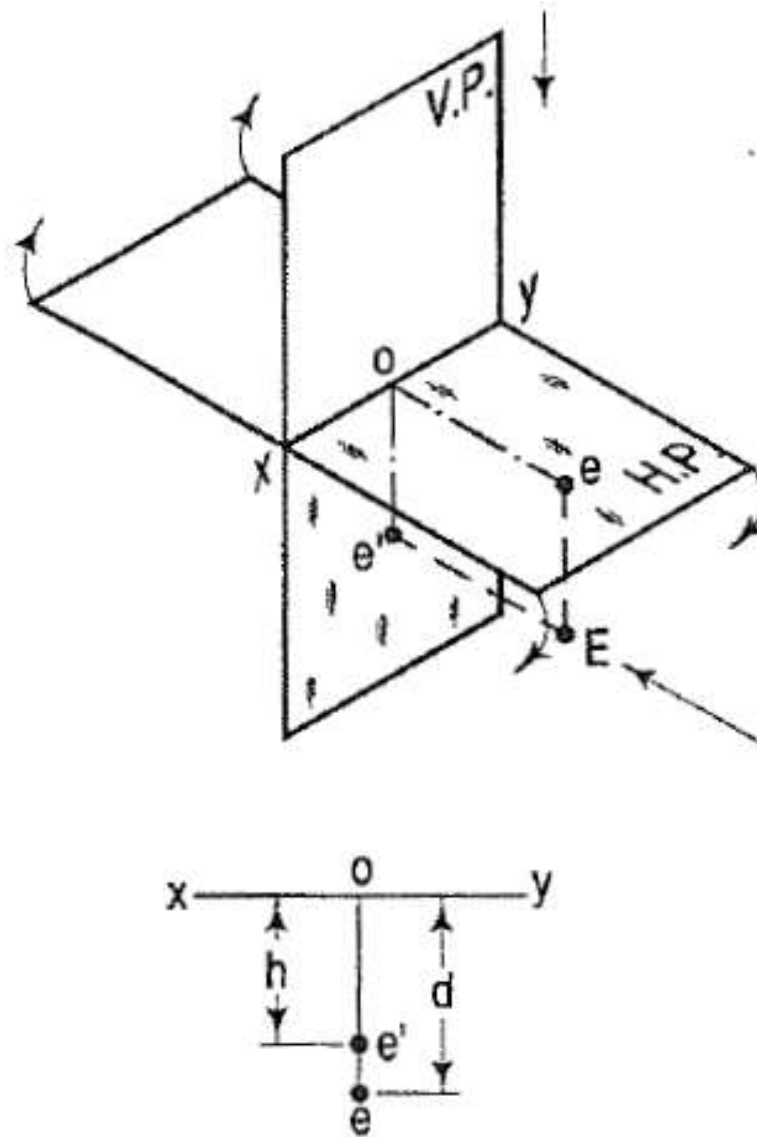
Projections of Points contd.

- 3. Point Situated in the third Quadrant



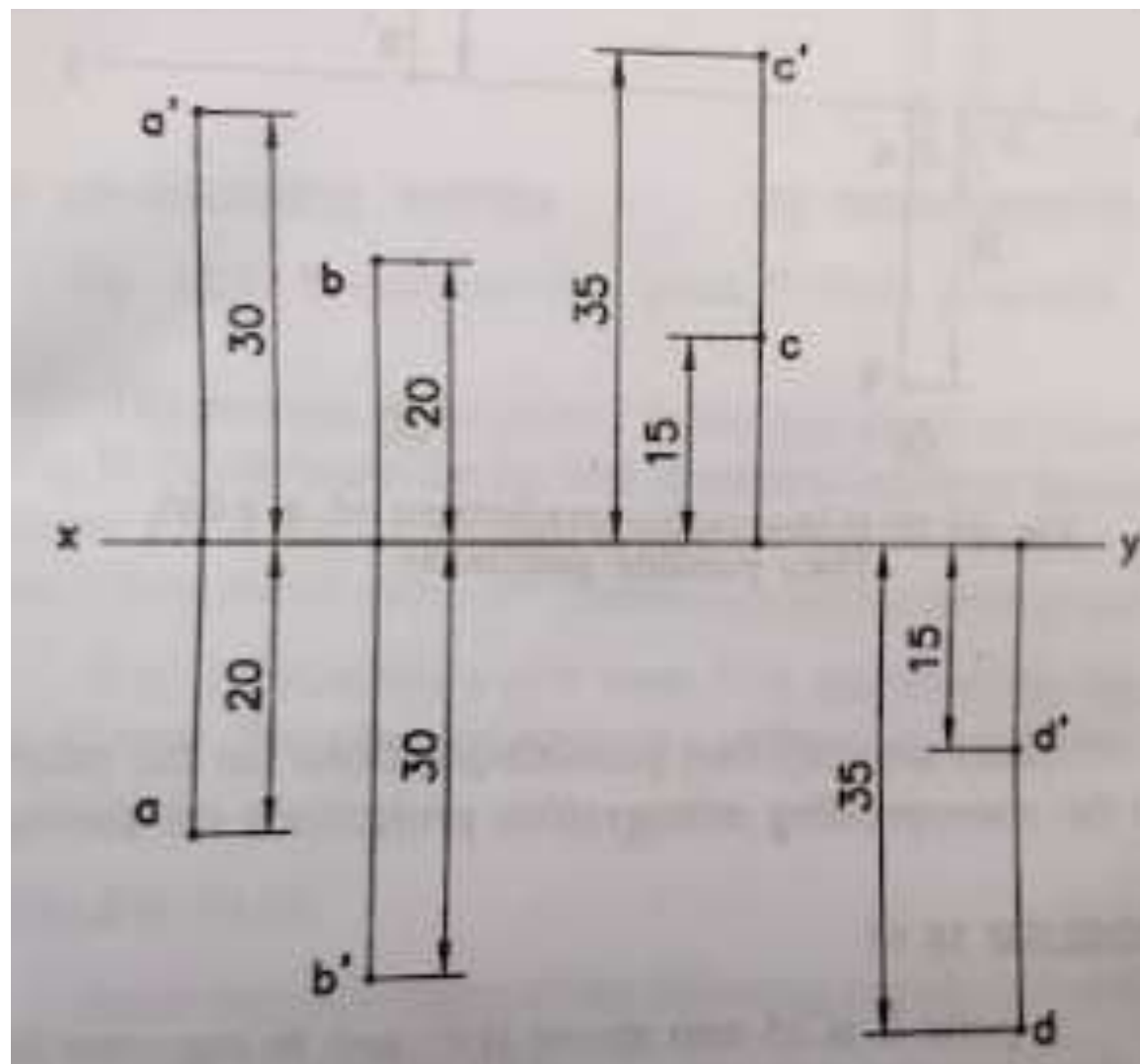
Projections of Points contd.

- 4. Point Situated in the fourth Quadrant



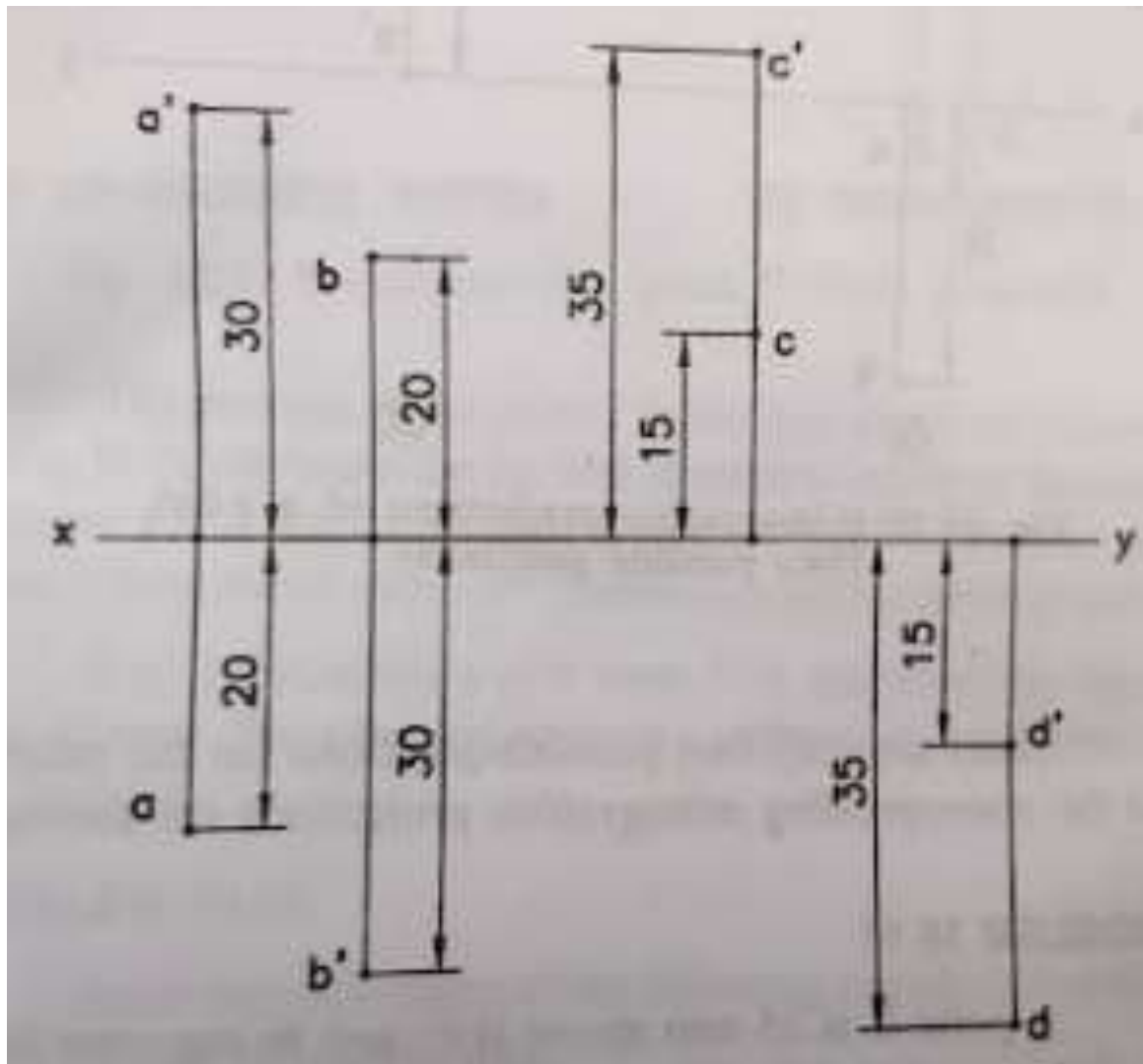
Projections of Points contd.

- **PRACTICE PROBLEM -1**
- Orthographic projection of A, B, C, and D are shown here. Read the views and state their projections w.r.t. HP and VP. Also state the quadrant in which they lie



Projections of Points contd.

- PRACTICE PROBLEM-1**



1. Point A is 30 mm above HP and 20 mm in front of VP- First Quadrant
2. Point B is 30 mm below HP and 20 mm behind VP- Third quadrant
3. Point C is 35 mm above HP and 15 mm behind VP- Second Quadrant
4. Point D is 15 mm below HP and 35 mm in front of VP- Fourth Quadrant

Projections of Lines

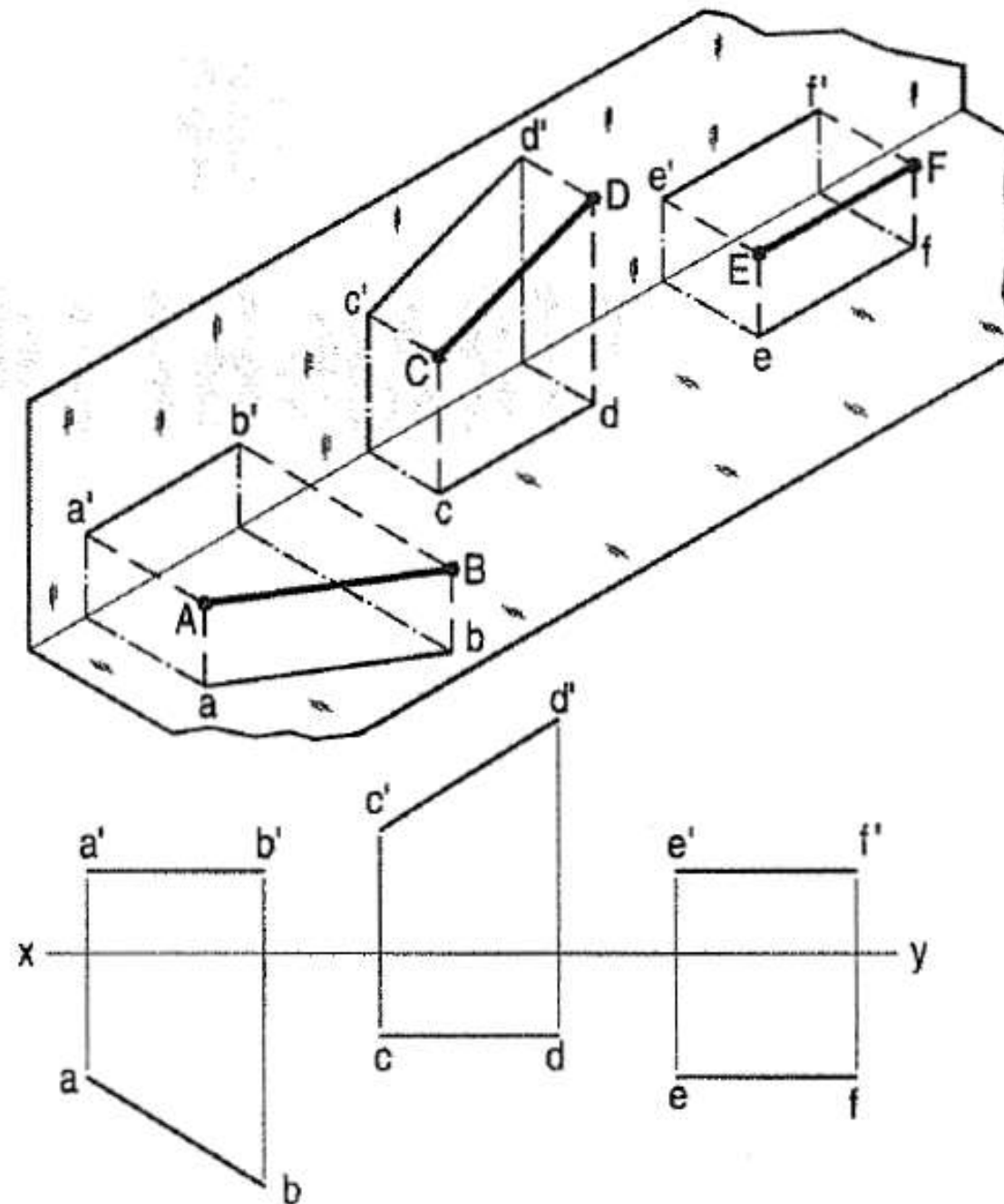
- Introduction
- Line Parallel to one or both planes
- Line contained by one or both planes
- Line perpendicular to one of the planes
- Line inclined to one of the planes and parallel to the other
- Line inclined to both the planes
- Traces of a line

Projections of Lines contd.

- **Introduction**
- A straight line is the shortest distance between two points. Hence the projections of a straight line may be drawn by joining the respective projections of its ends which are points

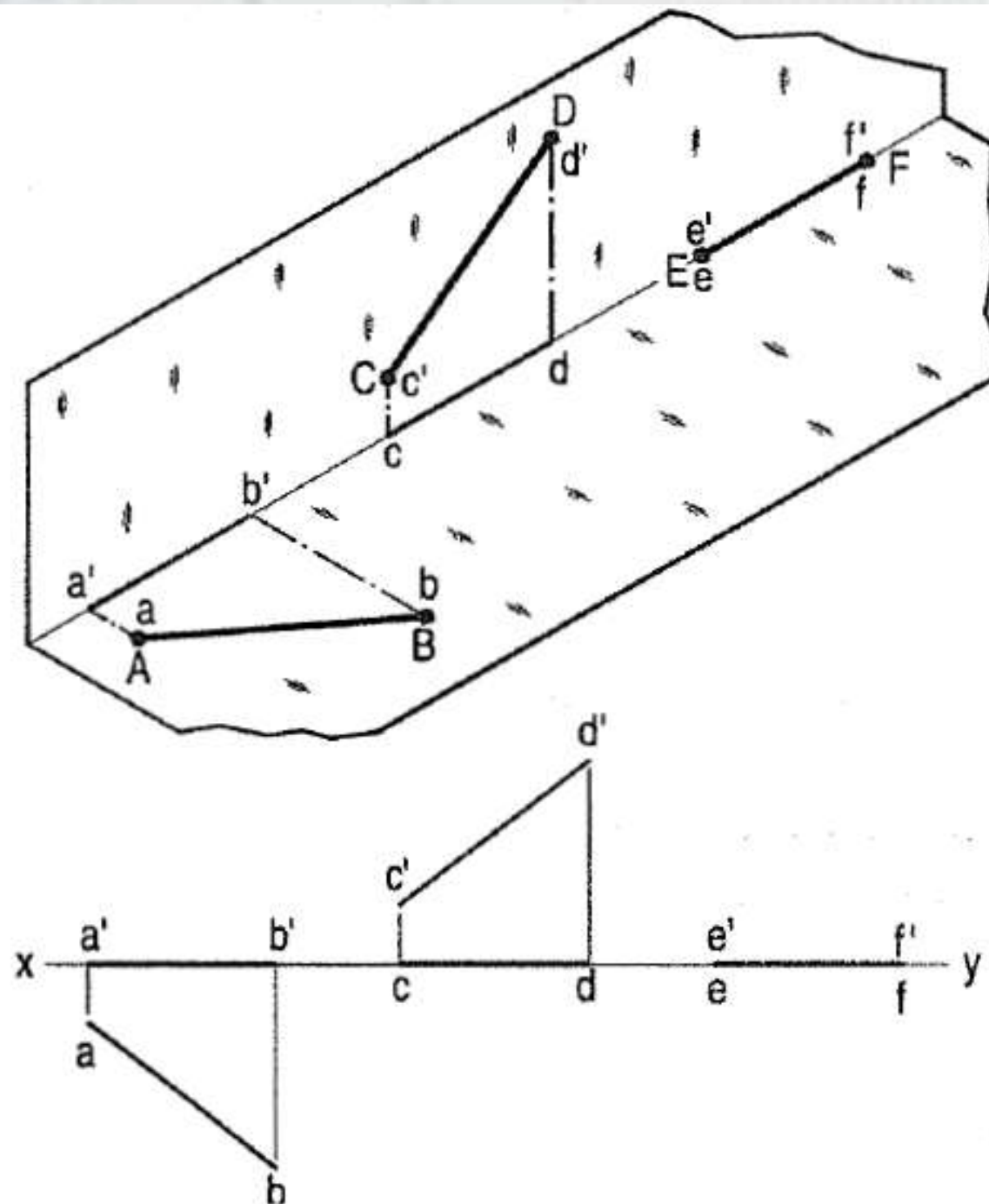
Projections of Lines contd.

- Line Parallel to one or both planes



Projections of Lines contd.

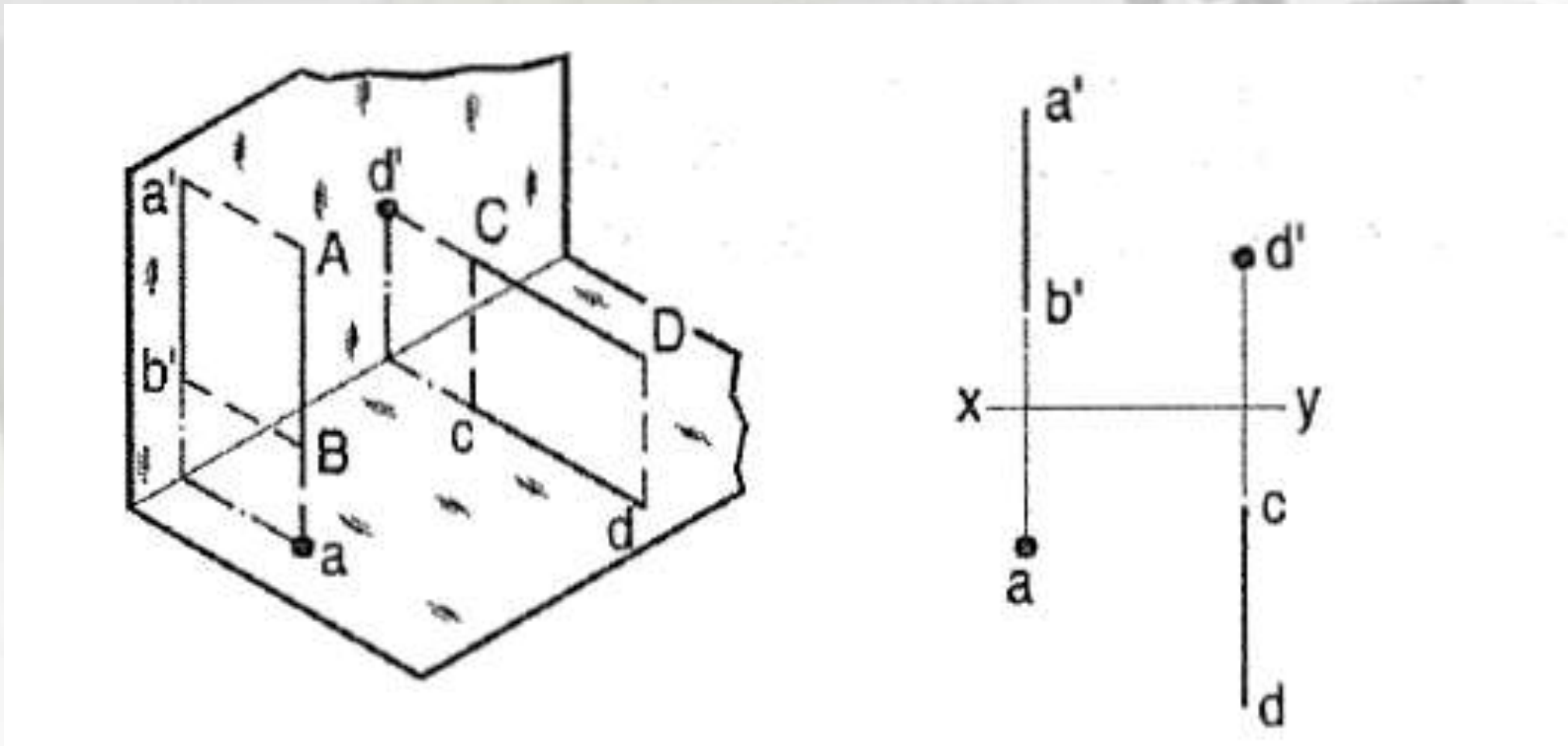
- Line contained by one or both planes



Source: N. D. Bhatt

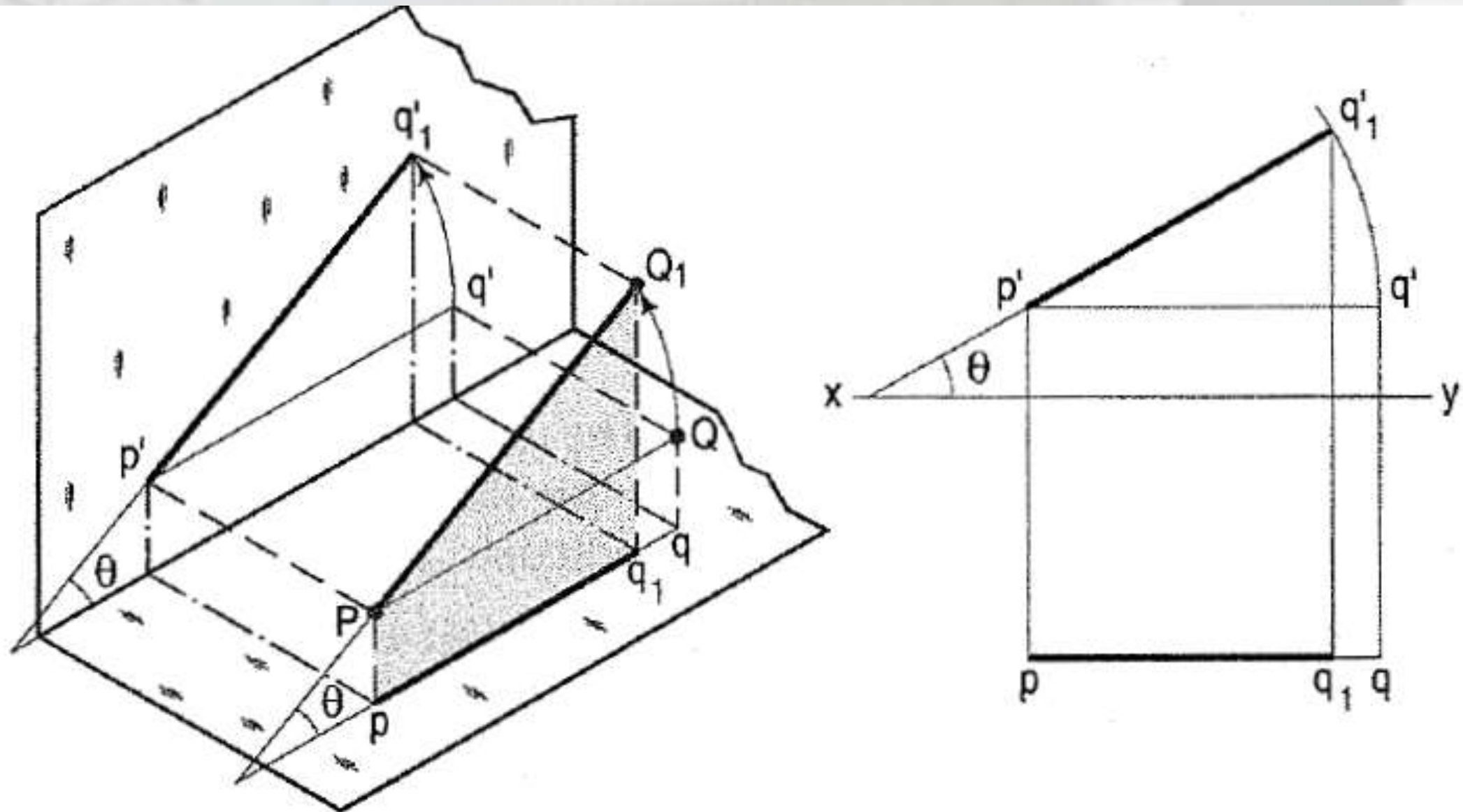
Projections of Lines contd.

- Line perpendicular to one of the planes



Projections of Lines contd.

- Line inclined to one plane and parallel to another



Projections of Lines contd.

• PRACTICE PROBLEM -2

Problem 10-2. (fig. 10-7): *The length of the top view of a line parallel to the V.P. and inclined at 45° to the H.P. is 50 mm. One end of the line is 12 mm above the H.P. and 25 mm in front of the V.P. Draw the projections of the line and determine its true length.*

As the line is parallel to the V.P., its top view will be parallel to xy and the front view will show its true length and the true inclination with the H.P.

- (i) Mark a , the top view, 25 mm below xy and a' , the front view, 12 mm above xy .
- (ii) Draw the top view ab 50 mm long and parallel to xy and draw a projector through b .
- (iii) From a' draw a line making 45° angle with xy and cutting the projector through b at b' . Then $a'b'$ is the front view and also the true length of the line.

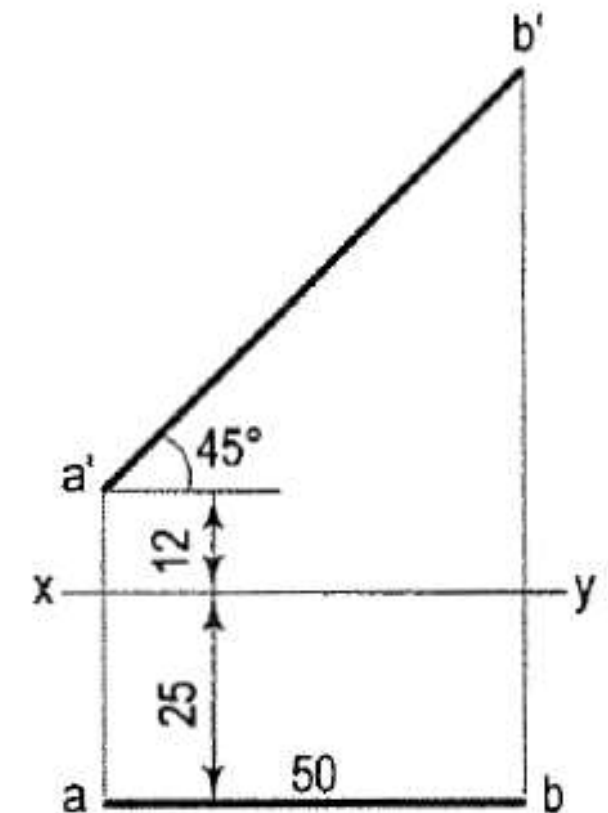
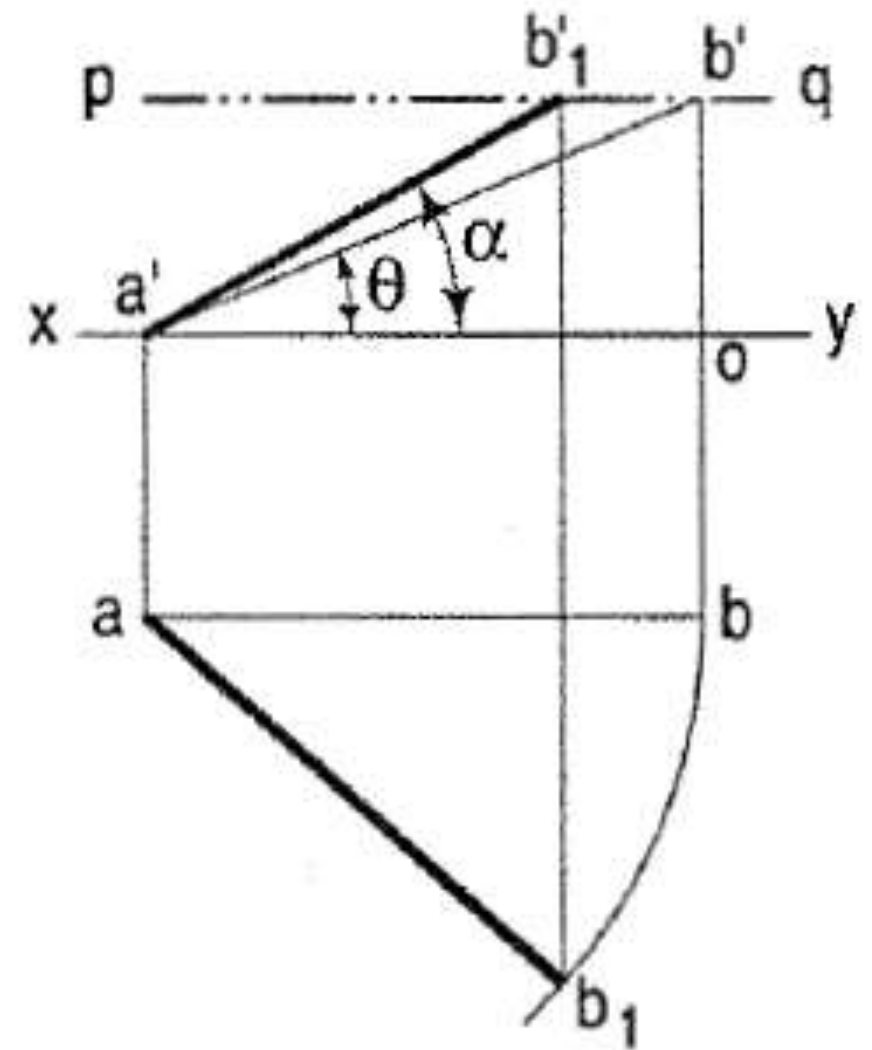
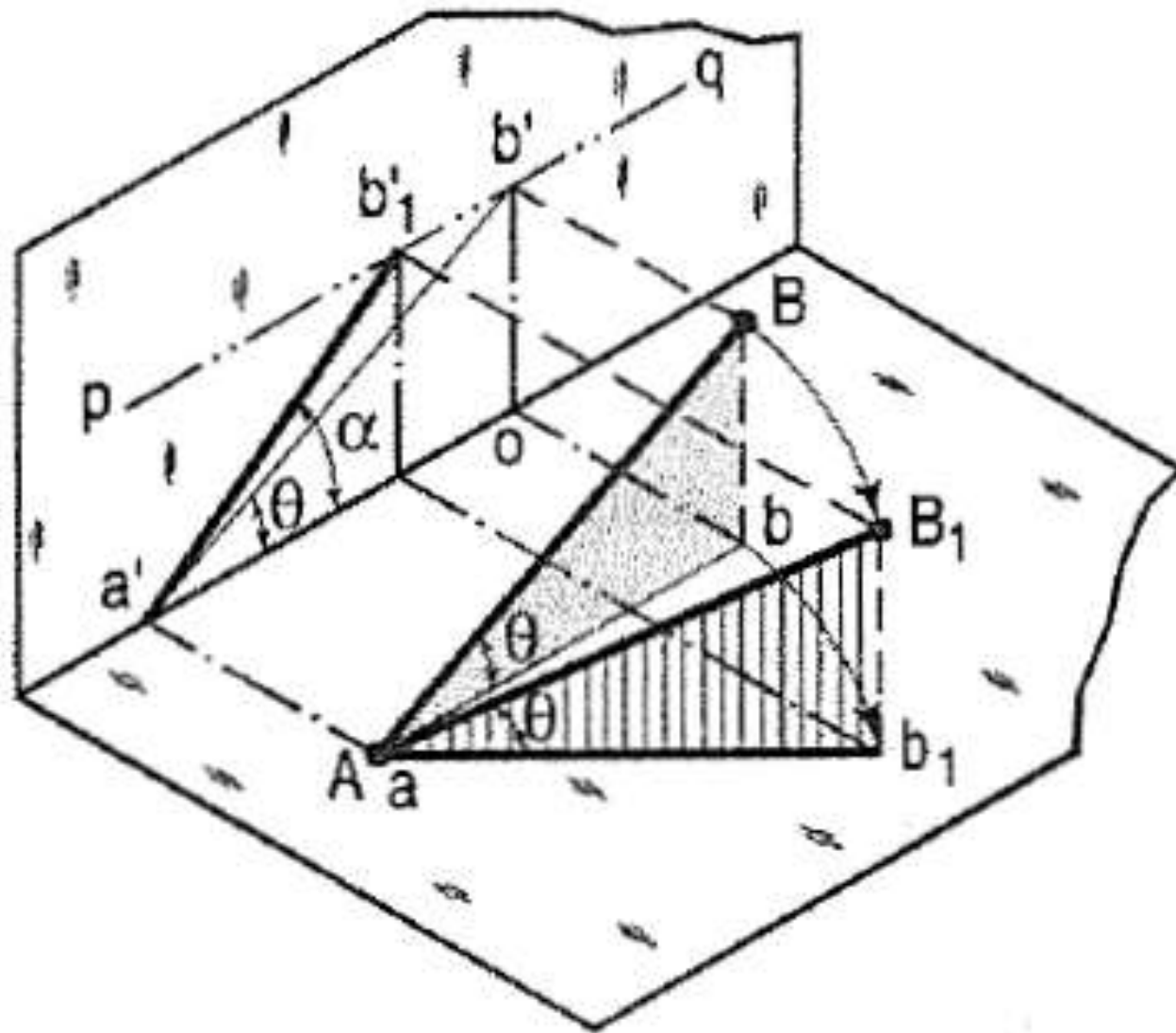


FIG. 10-7

Projections of Lines contd.

- Line inclined to both the planes



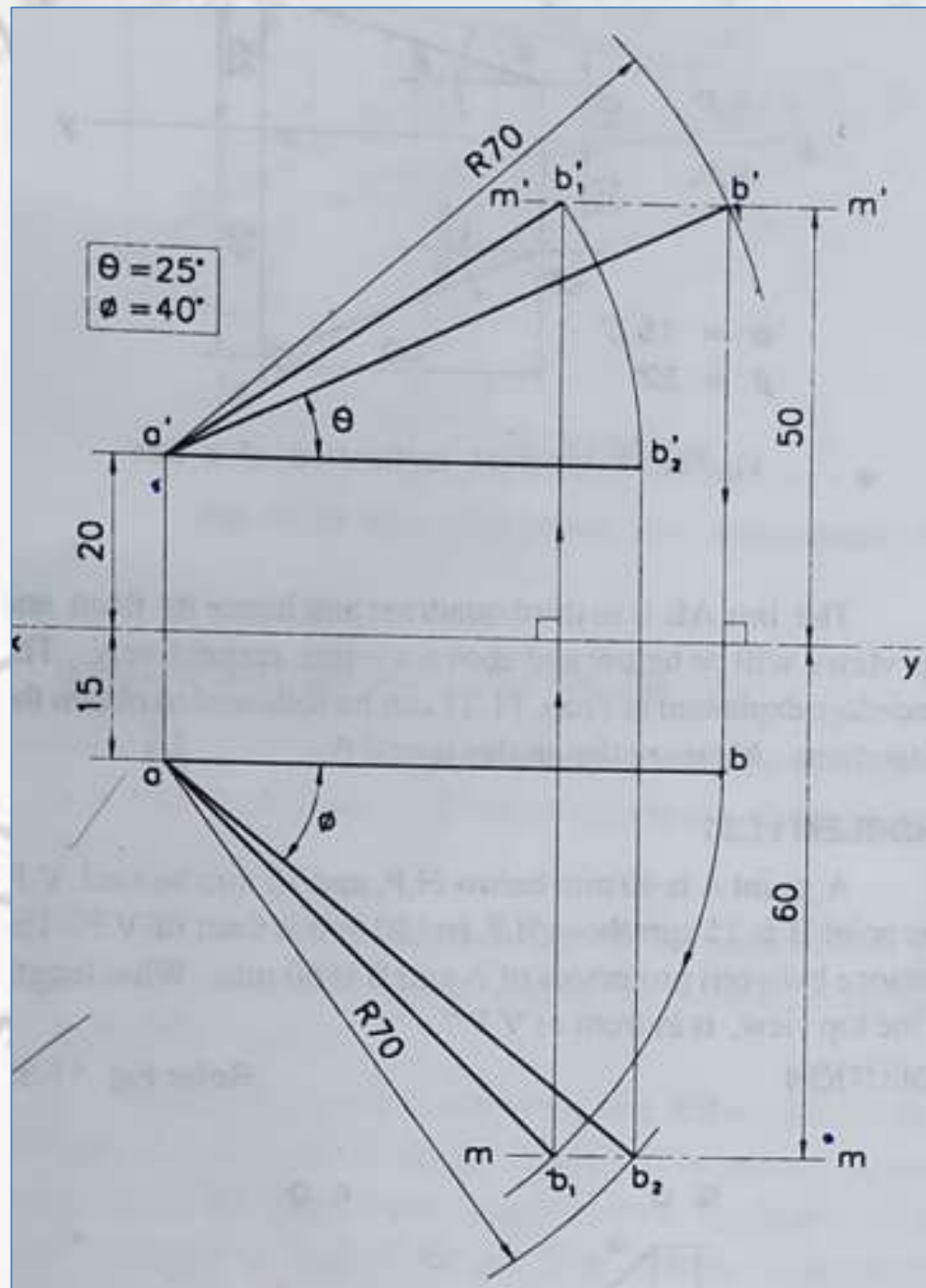
Projections of Lines contd.

- **PRACTICE PROBLEM -3**

A line AB measuring 70 mm has its end A 15 mm in front of VP and 20 mm above HP and other end B is 60 mm in front of VP and 50 mm above HP. Draw the projections of the line and find the inclinations of the line with both the reference planes of projections.
(KRP-'94)

Projections of Lines contd.

• PRACTICE PROBLEM -3



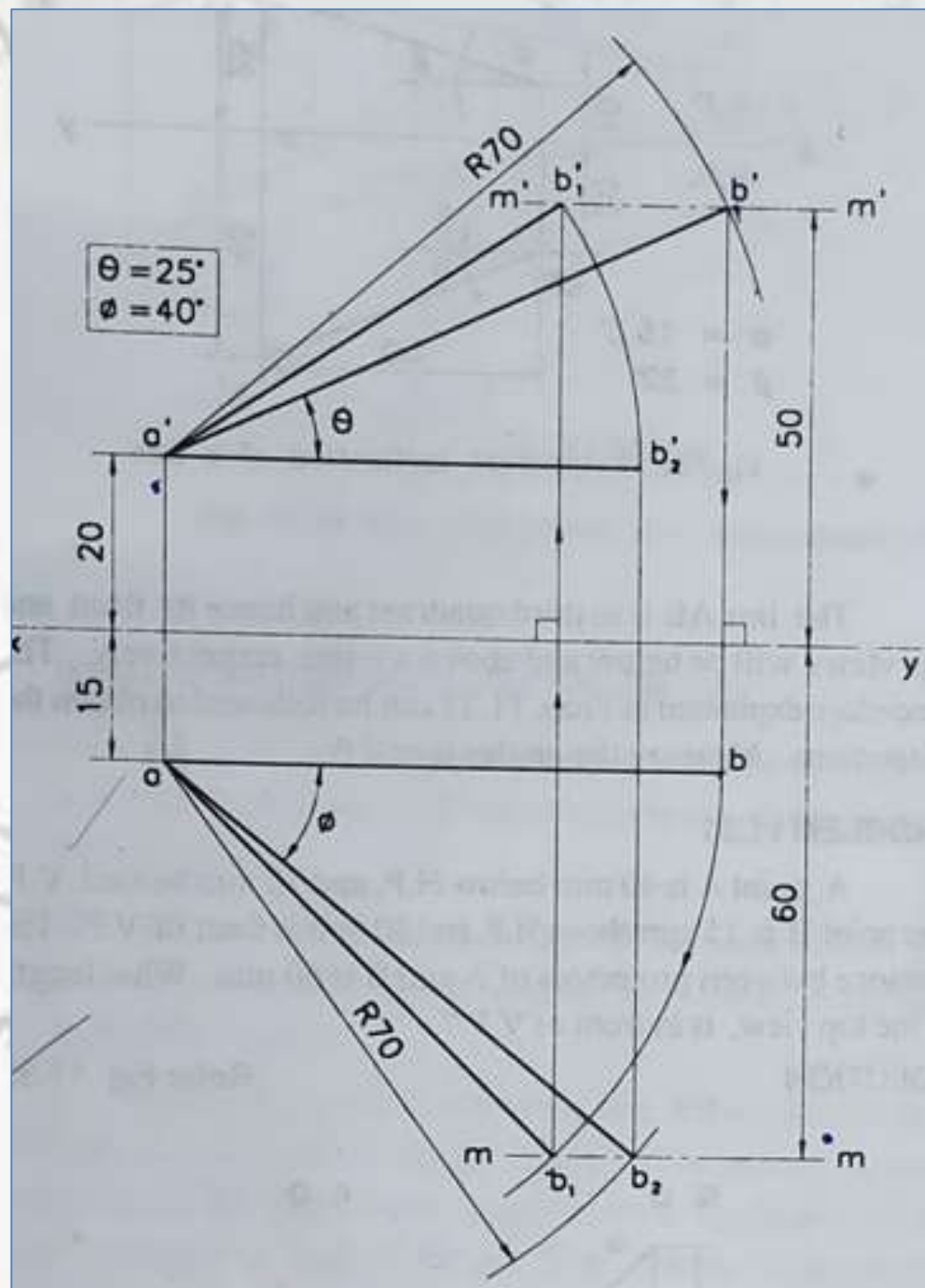
The line AB is in the first quadrant. Hence, its front view will be above xy-line and its top view will be below xy-line. The procedure is described below:

1. Draw the reference line xy.
2. Locate the points a' and a.
3. Locate the locus line mm of point B in the top view at a distance of 60 mm below xy-line and parallel to it.
4. Locate the locus line m'm' of point B in the front view at a distance of 50 above xy-line and parallel to it.
5. With centre a' and radius equal to 70 mm draw an arc to cut the locus line m'm' at b'.

Source: P. I. Varghese

Projections of Lines contd.

- PRACTICE PROBLEM -3**



6. Draw a vertical projector downwards through the point b' , to intersect the horizontal through a at b .

7. With centre a and radius equal to ab draw an arc to intersect the locus line mm at b_1 . Join ab_1 which represents the top view of the given line.

8. Draw a vertical projector through b_1 upwards to intersect the locus line of B in front view at b_1' . Join $a'b_1'$ which represents the front view of the given line.

9. To get the true inclination of the line AB with HP measure the angle θ of $a'b'$ with xy -line.

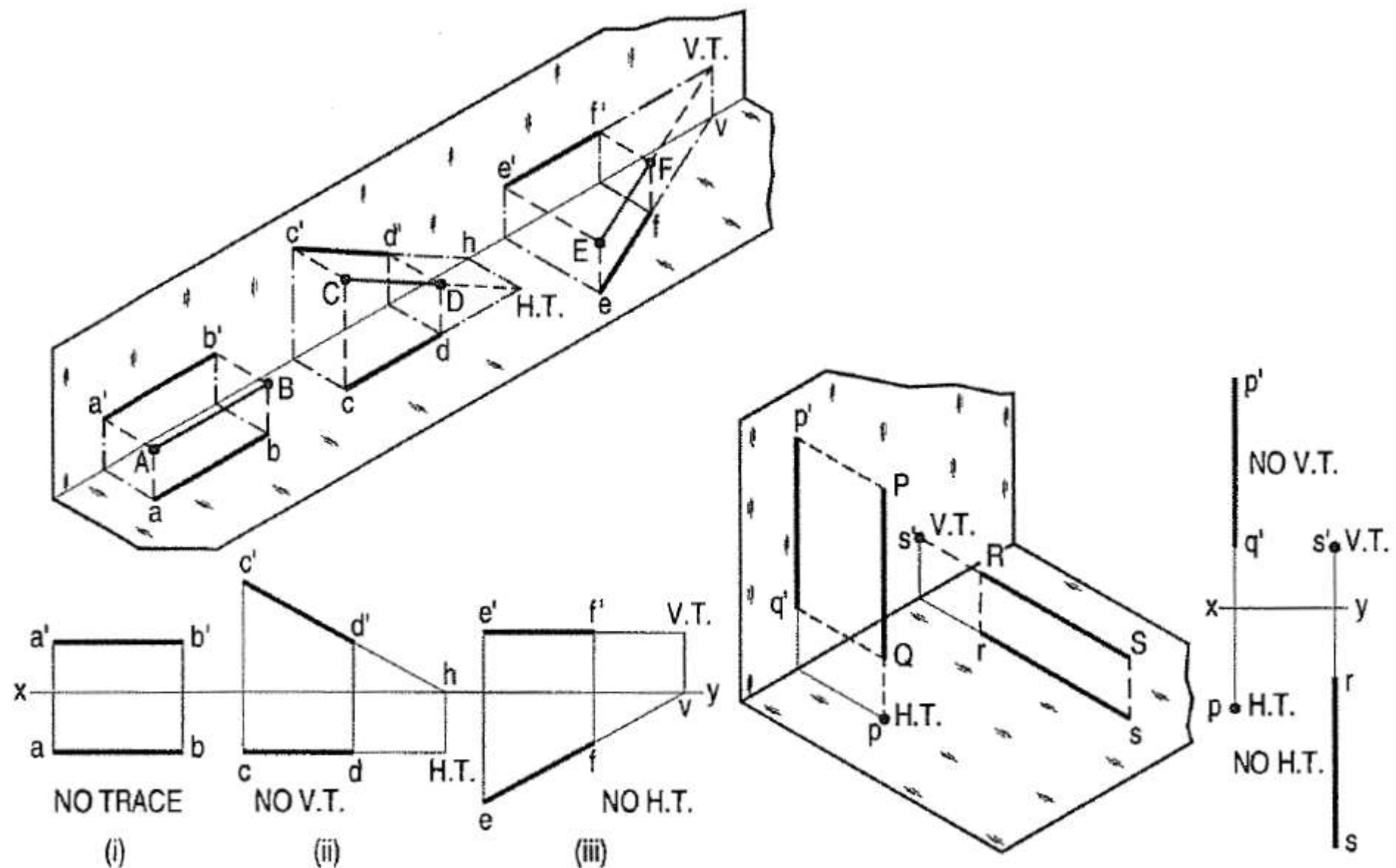
10. With centre a and radius equal to 70 mm draw an arc to cut the locus line mm at b_2 . Join ab_2 . To get the true inclination of the line AB with VP measure the angle ϕ of ab_2 with xy -line.

Source: P. I. Varghese

Projections of Lines contd.

- **Trace of a Line**

The point in which the line or line produced meet at the plane is called its trace



Projections of Lines contd.

- **Trace of a Line contd.**

There are two types of traces:

(1) Horizontal Trace

(2) (2) Vertical Trace

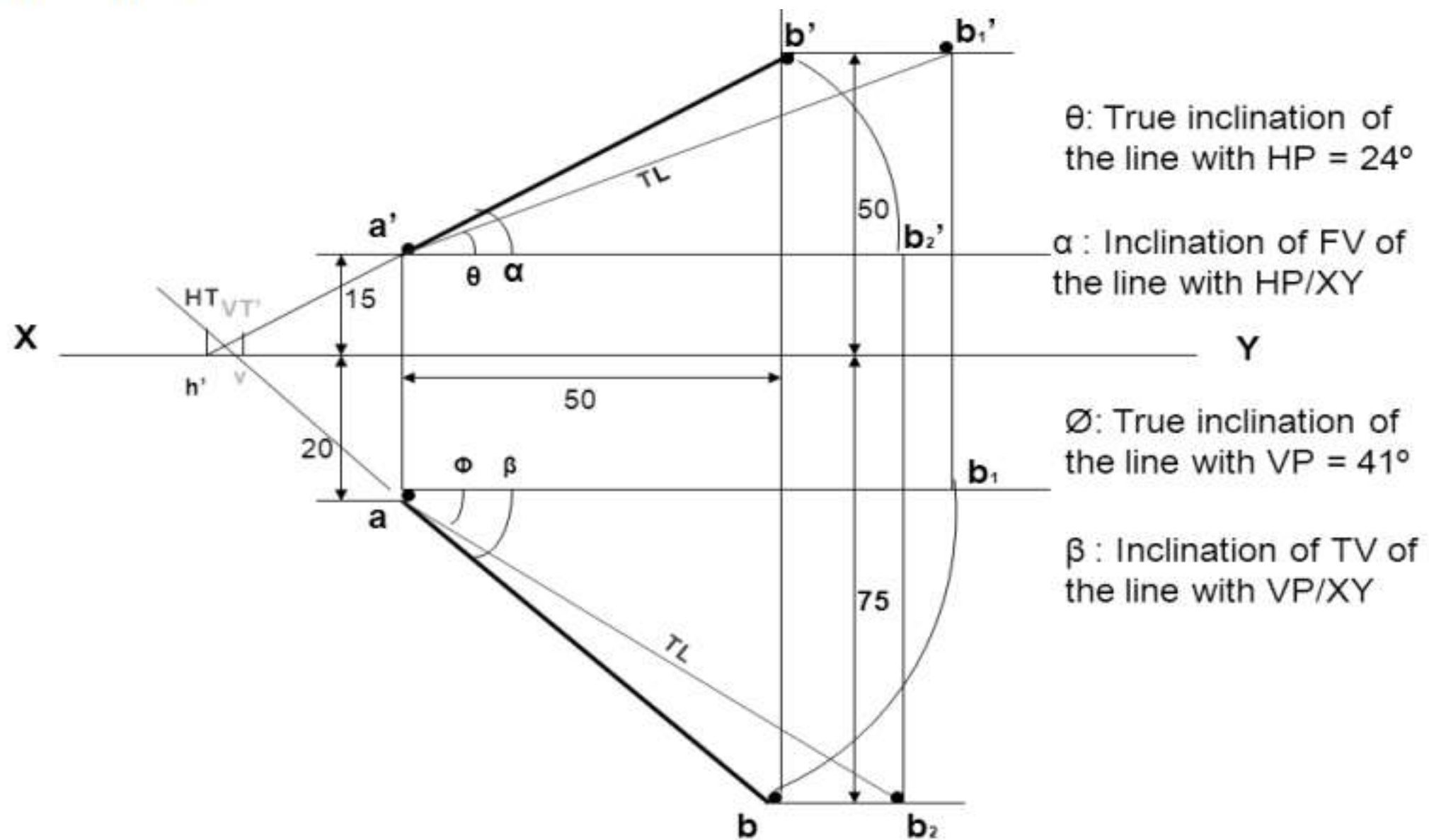
Horizontal Trace: When the front view line whilst extending intersects the reference line (xy) the point is mark as 'h' and a line is drawn from 'h' to intersect at top view extended line and mark as HT (h') which is known as horizontal trace

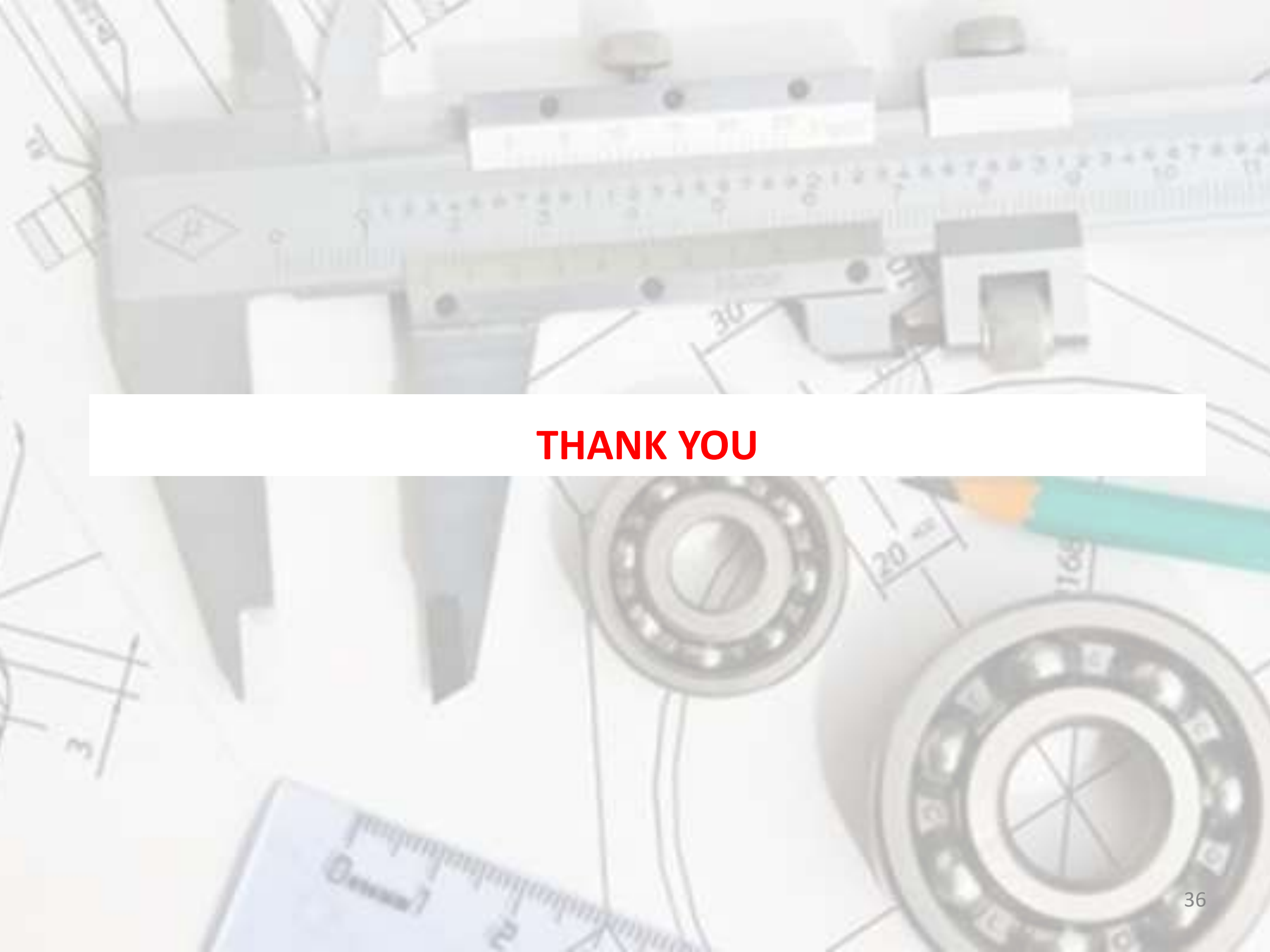
Vertical Trace: When the top view line whilst extending intersects the reference line (xy) the point is mark as 'v' and a line is drawn from 'v' to intersect at front view extended line and mark as VT (v') which is known as vertical trace

Projections of Lines contd.

• PRACTICE PROBLEM -4

End A of a line AB is 15mm above HP & 20mm in front of VP while its end B is 50mm above HP and 75mm in front of VP. The distance between end projectors of the line is 50mm. Draw projections of the line and find its true length and true inclination with the principal planes. Also mark its traces.



The background is a faded technical drawing of mechanical components. It includes a ruler at the top with millimeter markings, two ball bearings, a green pencil, and various dimension lines with numbers like 30, 20, 160, and 3. A diamond-shaped symbol with the letter 'H' is also visible.

THANK YOU